

WATTWISE · PHASE 2 · POST-BUILD & VALIDATION PLAYBOOK

All 8 sprints done. What now? And how do you test without a factory?

The build is complete. The code runs. Now the hard part: getting a you, validating ML predictions against real-world conditions, and the entire system without stepping foot inside a textile mill in Faisalabad.

✓ 8/8 Sprints Complete

5 Ways to Test Without a Factory

PHASE 2 PRIORITY MAP

PHASE 2 · STARTS NOW

What to Do Next — In This Exact Order

You've built the product. The next phase is **not more engineering**. It's validation, customer acquisition, and fundraising — with targeted engineering only where it proves value. Here's the priority stack.

1 Build a home test bench this week (Rs. 12,000–20,000)

You cannot wait for a factory. Build a physical lab in your room that simulates factory electrical conditions. Buy CT clamps, plug them on a power strip running your laptop, fan, and iron. Validate the full pipeline with real hardware right now. Full shopping list is in this guide.

2 Build the Digital Twin simulation engine (2 weeks of coding)

Upgrade your `factory_sim.py` from a random number generator into a full physics-based textile mill digital twin using real WAPDA/FESCO historical load shedding data. This becomes your primary testing environment.

3 Write a comprehensive test suite (2 weeks)

Unit tests, integration tests, ML backtests, chaos tests, and end-to-end tests. Until the test suite is green, the product is not production-ready regardless of what the code looks like.

4 Get a remote shadow monitoring agreement (next 4 weeks)

You don't need to be physically present. Convince one factory owner (via APTMA, LUMS NIC connections, or LinkedIn) to let you install a Rs. 0 monitoring kit — one RPi CM4 + 4G SIM — and you observe remotely. No automation. No liability. Just data. This is how you get 30 days of real sensor data without being on-site.

5 Apply for LUMS NIC / PITB / Invest2Innovate incubation

These programs give you: a credible launch platform, connections to textile industrialists, legal help for customer agreements, and access to the APTMA industrial network. Apply this week — acceptance timelines are 4–8 weeks.

6

Prepare seed deck and financial model

Target: USD 200–300K seed from local angels. Your documentation and codebase are already investor-grade. You need a 12–slide pitch deck, a live demo video of the dashboard, and the savings calculator showing unit economics. Spend 2 weeks on this in parallel with testing.

PRIORITY MAP

The Phase 2 Priority Stack

Priority	Activity	Timeline	Why Now	Status
P0	Home test bench setup	This week	You can't validate anything without hardware	TODO
P0	Digital twin upgrade	Week 1–2	Primary test environment for all ML models	TODO
P0	Full test suite	Week 2–3	Nothing ships without tests. Non-negotiable.	TODO
P1	Shadow monitoring agreement	Week 2–4	Real data unlocks real ML model training	TODO
P1	WAPDA data integration	Week 2–3	GOP model needs historical feeder schedules	TODO
P1	Incubator applications	This week	4–8 week process — apply immediately	TODO
P2	Seed pitch deck	Week 3–4	You have the story and the code. Just write it.	TODO
P2	First customer agreement draft	Week 3	Legal review takes 2–4 weeks via lawyer	TODO
P2	PSQCA hardware certification	Week 4+	Required before commercial deployment	TODO



Production deployment
(AWS)

Already
done

Infra is live from Sprint 6–7



DONE

TESTING CHALLENGE

The Testing Problem — And Why It's Solvable

You cannot walk into a textile mill in Faisalabad tomorrow and say "let me test my software on your loom." They'll say no — reasonably. A factory is a live revenue-generating operation and an unknown startup's software near their ATS switch is a liability. This is a real constraint. But it's not a blocker.



The core insight: You don't need a factory to validate WattWise. You need *equivalent electrical signals*. A Rs. 15,000 home test bench produces exactly the same voltage readings, frequency sags, and current waveforms that a factory does — just at household scale instead of industrial scale. The CT sensor doesn't know if it's on a loom or a room heater.

5 Testing Methods — From Free to Field

#	Method	Cost	What It Validates	Factory Access Needed
M1	Automated Software Test Suite	Rs. 0	API, ML logic, billing calculations, data pipeline	None
M2	Physical Home Lab (Test Bench)	Rs. 12–20K	Full hardware stack: CT sensors, WattBrain, relay control	None
M3	Digital Twin Simulation	Rs. 0 (code)	ML models, load shift optimizer, outage predictions	None
M4	Remote Shadow Pilot	Rs. 5–10K hardware	Real sensor data, WAPDA outage patterns, production readiness	Remote only (4G)
M5	Chaos Engineering	Rs. 0	Failure modes: network drops, sensor fail, relay fault	None

Software Test Suite — Validate Everything Without Hardware

A

Go API — Integration Tests

Rs. 0

Test all endpoints: auth, tenant isolation, telemetry ingestion, billing

testify + docker

apps/api/internal/auth/auth_test.go

```
func TestCrossTenantAccessBlocked(t *testing.T) {
    // Factory A user tries to access Factory B data
    tokenA := login(t, "manager@millA.pk", "password")
    factoryB_ID := "fsd_mill_002"

    resp := apiGet(t, "/v1/factories/"+factoryB_ID+"/telemetry/live", tokenA)
    assert.Equal(t, 403, resp.StatusCode) // MUST be blocked
}

func TestSavingsLedgerImmutability(t *testing.T) {
    // Attempt to UPDATE savings record — must fail silently
    record := insertSavingsRecord(t, "fsd_mill_001", "2026-09", 5260000)
    err := db.Exec("UPDATE savings_records SET gross_saving_pkr=999 WHERE id=$1", record)

    // PostgreSQL Rule should silently prevent it
    fresh := getSavingsRecord(t, record.ID)
    assert.Equal(t, 5260000.0, fresh.GrossSavingPKR) // unchanged
}
```

- ☐ **Auth tests:** login, bad password, expired token, token refresh, logout
- ☐ **Tenant isolation:** cross-factory access blocked (403), JWT factory_ids list enforced
- ☐ **Savings immutability:** UPDATE/DELETE on savings_records silently blocked by PostgreSQL rules
- ☐ **MQTT ingestion:** valid packet accepted, out-of-range values rejected, duplicate packets idempotent
- ☐ **Invoice calculation:** 20% gain-share computed correctly on test savings amounts

B

ML Model — Backtesting on Historical Data

Rs. 0

Validate GOP, LoadShift, and Prophet against real Pakistani feeder data

NEPRA public data

NEPRA publishes daily generation and load shedding data. FESCO/LESCO publish feeder schedules. Use

this as your training AND testing dataset.

apps/ml/tests/test_gop_backtest.py

```
import pandas as pd
from sklearn.metrics import precision_score, recall_score, confusion_matrix

def test_gop_on_fesco_2025_data():
    # Load 12 months of real FESCO feeder logs (NEPRA Annual Report data)
    df = pd.read_csv("tests/data/fesco_feeder_A11_2025_actual.csv")

    # Run GOP predictions on this historical data
    y_true = df['actual_outage_occurred']
    y_pred = [gop_model.predict(row) > 0.85 for _, row in df.iterrows()]

    precision = precision_score(y_true, y_pred)
    recall = recall_score(y_true, y_pred)

    # MUST meet our advertised performance guarantees
    assert precision > 0.88, f"Precision {precision:.3f} below 88% SLA"
    assert recall > 0.80, f"Recall {recall:.3f} below 80% SLA"

    # False alarm rate: generator starts unnecessarily
    tn, fp, fn, tp = confusion_matrix(y_true, y_pred).ravel()
    false_alarm_rate = fp / (fp + tn)
    assert false_alarm_rate < 0.05, "Too many false generator starts!"
```



Where to get real data: NEPRA Annual Reports at nepra.org.pk (free, public). FESCO load shedding notices at fesco.com.pk. WAPDA national generation data via daily press releases. Compile 12 months of feeder data manually for the first version.

C

End-to-End Tests with Playwright

Test the full UI flow: login → dashboard → savings certificate download

Rs. 0

playwright

apps/web/tests/e2e/dashboard.spec.ts

```
import { test, expect } from '@playwright/test'

test('factory manager sees live telemetry dashboard', async ({ page }) => {
    // Login as factory manager (test tenant)
    await page.goto('http://localhost:5173/login')
    await page.fill('[data-testid=email]', 'manager@test-mill.wattwise.pk')
```

```

await page.fill('[data-testid=password]', process.env.TEST_PASS)
await page.click('[data-testid=login-btn]')

// Dashboard should load with live power readings
await expect(page.locator('[data-testid=total-kw]')).toBeVisible()
await expect(page.locator('[data-testid=grid-status]')).toContainText(/(ON|GENERATOR)/

// Savings certificate download should work
const download = await page.waitForEvent('download', async () => {
  await page.click('[data-testid=download-esco-cert]')
})
expect(download.suggestedFilename()).toMatch(/WattWise_ESCO.*\.pdf/)
})

```

METHOD 2 · RS. 12,000-20,000 · MOST IMPORTANT

Physical Home Lab — A Rs. 15,000 Factory Substitute

This is the most important step you can take this week. Your RPi CM4 and CT sensors don't know if they're in a textile mill or your bedroom. Build this bench today and you have a permanent hardware validation environment.

WATTWISE HOME TEST BENCH — PHYSICAL LAYOUT

CT Clamp × 3

On power strip cables



RS-485 Bus

USB adapter to RPi



RPi CM4

WattBrain OS



MQTT/TLS

Via WiFi to cloud



Dashboard

wattwise.pk

Laptop

"Weaving Shed"

~65W base

Room Heater (1kW)

"Dyeing Vat"

resistive load

Fan (75W)

"HVAC"

inductive load

Power Drill (300W)

"CNC Machine"

motor load

Smart Plug

Relay substitute

RS-485 controlled

Plug all loads into a multi-socket power strip. Put CT clamps on the individual cables feeding each socket. Control the smart plug as your simulated ATS relay.

What This Tests — Seriously

Hardware Validated

- CT sensor reading accuracy (verify against phone clamp meter)
- Modbus RS-485 data bus reliability
- WattBrain ONNX inference latency
- Relay response time (smart plug switching)
- 72h SQLite offline buffer
- OTA update across WiFi

Software Validated

- Full MQTT → Kafka → InfluxDB pipeline with real hardware
- WebSocket updates appearing in dashboard
- Load changes (turning heater on/off) reflected in real-time
- Anomaly detector triggering on simulated faults
- Power factor calculation on inductive fan load

Simulate Factory Events at Home

Factory Event	Home Simulation	What to Verify
Grid load shedding (full outage)	Pull the main power strip plug for 30 seconds	SQLite buffer captures gap; data reconciles when reconnected
Generator switchover (ATS)	Toggle smart plug OFF then ON via relay command	<8ms relay response; load returns; no data gap in InfluxDB
High-load machine start (loom)	Plug in 1kW heater suddenly	Anomaly detector sees load spike; no false positive alert
CT sensor tamper / disconnect	Remove one CT clamp mid-session	Anomaly detector fires within 90 seconds; alert sent via WhatsApp
4G network drop (offline mode)	Disable RPi WiFi for 2 hours	All data buffered in SQLite; uploads cleanly when WiFi restored
Voltage sag (grid instability)	Run power drill at half-speed (dimmer)	Frequency/voltage sensors catch the sag; GOP confidence rises
Power factor degradation	Run fan on AC through variac	PF reading drops below 0.85; alert fires

[HARDWARE SHOPPING LIST](#) · [BUY TODAY](#)

Test Bench Shopping List

Everything available at Hafeez Centre (Lahore), Raja Bazaar electronics (Rawalpindi), or Aliexpress/Daraz with 3–7 day delivery.

Raspberry Pi CM4 (4GB RAM, 32GB eMMC) + IO Board

Rs. 12,000

Aliexpress / Hafeez Centre Lahore

Already ordered?

Split-core CT Current Clamps × 5 (SCT-013-100, 100A)

Rs. 3,500

Aliexpress (~\$3 each) or Hafeez Centre

USB-to-RS485 converter (CH340 chip, 3.3V/5V)

Rs. 500

Daraz.pk / Hafeez Centre

RS-485 TTL Modbus module × 2 (MAX485 chip)

Rs. 400

Daraz.pk electronics components

WiFi Smart Plug (Tuya-compatible) × 2

Rs. 1,800

Daraz.pk / any electronics shop

1kW Electric Convection Heater (simulates resistive load)

Rs. 2,500

Any electronics/appliance shop

Variac / Variable Transformer 0–240V (for voltage sag simulation)

Rs. 4,000

Raja Bazaar Rawalpindi / electrical wholesale

Clamp meter (for ground truth verification of CT readings)

Rs. 3,500

Any hardware store (UNI-T UT210E)

4G USB Dongle (Telenor or Jazz) + SIM

Rs. 2,000

Any telecom outlet

Patch panel / DIN rail enclosure for professional mounting

Rs. 1,500

Electrical wholesale market

TOTAL BENCH COST

Rs. ~32,000

Less Rs. 12K if CM4 already ordered = Rs. 20K

read it via an ADS1115 ADC chip connected to RPi GPIO pins — this is the exact circuit inside a real WattClamp™ node. You're not approximating factory conditions. You're running the same hardware.

METHOD 3 · FREE · 2 WEEKS OF CODING

Digital Twin — A Physics-Accurate Textile Mill in Python

Upgrade your `factory_sim.py` from a random number generator into a factory that behaves like a real mill, including realistic load curves, Pakistani load shedding patterns, and dye vat thermal dynamics.

`apps/edge/simulator/digital_twin.py` — Physics-accurate mill simulator

```
import pandas as pd, numpy as np
from datetime import datetime, timedelta
import json, time, paho.mqtt.client as mqtt

# === REAL FESCO LOAD SHEDDING SCHEDULE (scraped from wapda.gov.pk) ===
# Feeder A-11, Faisalabad Industrial Zone - Historical 2025 Data
FESCO_A11_SCHEDULE = pd.read_csv("data/fesco_feeder_A11_schedule_2025.csv")

# === TEXTILE MILL PROCESS MODEL ===
class DyeingVat:
    """High-temperature pressurized dye vat — 130°C batch cycle"""
    def __init__(self, name, base_kw=71):
        self.name = name
        self.base_kw = base_kw
        self.temp = 25.0          # °C — ambient start
        self.state = 'IDLE'       # IDLE | HEATING | DYEING | COOLING
        self.cycle_min = 0

    def step(self, dt_min=1, power_available=True):
        if self.state == 'HEATING':
            if power_available:
                self.temp += 0.8 * dt_min # 0.8°C/min heating rate
                if self.temp >= 130: self.state = 'DYEING'
            else:
                self.temp -= 0.3 * dt_min # cooling without power
                # CRITICAL: if temp drops below 118°C during dye phase = RUINED
                if self.temp < 118 and self.state == 'DYEING':
                    self.state = 'RUINED'

        kw = self.base_kw if power_available and self.state != 'IDLE' else 2.1
        return {'kw': kw, 'temp': self.temp, 'state': self.state}
```

```

class TextileMill:
    """Full Faisalabad-style textile mill - 40 looms + 4 dye vats"""
    def __init__(self, factory_id):
        self.factory_id = factory_id
        self.loods = [{'rpm': 0, 'base_kw': 7.2} for _ in range(40)]
        self.vats = [DyeingVat(f"VAT_{i+1}") for i in range(4)]
        self.on_grid = True
        self.gen_started = False

    def get_grid_status(self, timestamp):
        # Use REAL FESCO historical schedule to determine grid state
        hour = timestamp.hour
        feeder_row = FESCO_A11_SCHEDULE[FESCO_A11_SCHEDULE['hour'] == hour]
        outage_prob = feeder_row['outage_probability'].values[0]
        return np.random.random() > outage_prob

    def tick(self, ts):
        self.on_grid = self.get_grid_status(ts)
        readings = []

        # Loom section - total ~288 kW when all 40 running
        loom_kw = sum(7.2 * (1.0 if self.on_grid else 0.0) for l in self.loods)
        readings.append({'node_id': 'weaving_a', 'power_kw': loom_kw + np.random.normal(0,8)})

        # Dye vats - physics model
        for v in self.vats:
            r = v.step(power_available=self.on_grid)
            readings.append({'node_id': v.name, 'power_kw': r['kw'],
                             'temp_c': r['temp'], 'state': r['state']})

        return readings

```



Key insight: The DyeingVat class with thermal physics is how you validate that WattWise actually prevents the Rs. 450,000 dye batch loss. If SwiftSwitch™ triggers before grid drops, vat temp stays above 118°C. If it doesn't, the state becomes RUINED. This is your most compelling demo.

WAPDA/FESCO Data Integration

```

$ # Download 2 years of FESCO published outage data
$ curl -O https://fesco.com.pk/load-management/schedule-2025.xlsx

$ # Also check NEPRA Annual Report PDFs for grid-level data
$ # nepra.org.pk → Publications → Annual Reports → Download 2024-25
$ # Extract Table 4.3 "Load Shedding by DISCO" → CSV → training data

```

METHOD 4 · RS. 5-10K · BIGGEST IMPACT

Remote Shadow Pilot — Real Factory, No Physical Presence

This is the bridge between lab testing and full deployment. Convince one factory to let you install a **monitoring-only** kit — absolutely no automation, no relay control, no ATS connection. Just sensors watching. You observe from Islamabad via 4G.



The pitch that works: "We'd like to install Rs. 0 worth of non-invasive clip-on sensors on your electrical panel for 30 days. No rewiring, no downtime, no automation — just monitoring. At the end of 30 days, we'll give you a free report showing how much money you're losing to load shedding, with exact numbers. You keep the report regardless." No rational factory owner says no to free data.

Remote Shadow Kit

Hardware (Rs. 8,000 per site)

- RPi CM4 (already have some)
- 5-8 CT clamps (100A)
- Jazz 4G USB dongle + unlimited SIM (Rs. 1,200/mo)
- Small 5V power adapter for RPi
- Cable ties + DIN rail

Install time: 45 minutes by yourself.

What You Get

- 30 days of real industrial consumption data
- Actual WAPDA outage times on that feeder
- Real load curves for your ML training
- GOPmodel validation against actual outages
- Prophet baseline on real consumption
- **Proof your hardware works in an industrial environment**

How to Get Shadow Access — 4 Channels

- ☐ **LUMS NIC / PIEAS network:** Talk to your faculty advisors. Every engineering professor in Pakistan knows a factory owner. Ask for one introduction. One is enough.

- ☐ **APTMA** aptma.org.pk lists member mills with contact details. Email 20 mill owners with a 3-line pitch. Offer **website:** the free 30-day monitoring report. Expect 2–3 responses.
- ☐ **LinkedIn direct outreach:** Search "Factory Manager Faisalabad" or "GM Operations Textile" on LinkedIn Pakistan. Send personalized connection requests. 10 conversations → 1 pilot.
- ☐ **Family / friend network:** In Pakistan, someone you know almost certainly has a contact in textile manufacturing. Ask everyone. The industrial and educational networks overlap more than you think.

Remote Monitoring Architecture (No Physical Presence)

```
# WattBrain in factory (Faisalabad) —[Jazz 4G]—▶ mqtt.wattwise.pk —▶ Your Dashboard
#
# You SSH into WattBrain remotely if needed:
ssh -p 22 wattbrain@[dynamic-4G-IP] # Via Cloudflare Tunnel (no static IP needed)

# Or use Cloudflare Tunnel for permanent remote access:
cloudflared tunnel --url tcp://localhost:22 # Runs on WattBrain, free tier
# Factory owner never needs to open any firewall ports.
# You SSH in via: ssh username@tunnel-id.cfargotunnel.com
```



Trust-building detail: Show the factory owner that the WattBrain has *no relay outputs connected* during shadow mode. Physically show them the unit — 4 wires going to CT clamps (read-only), 1 wire for 5V power, 1 USB 4G stick. That's it. Nothing connected to their ATS. Nothing that can switch anything. This completely eliminates any worry about your software touching their production.

METHOD 5 • FREE • PRODUCTION READINESS

Chaos Engineering — Break It Before the Factory Does

Chaos engineering deliberately induces failures to find weaknesses before production. For WattWise this is especially critical — a software crash during a factory switchover could damage equipment or injure workers. Find every failure mode in the lab.

infra/chaos/scenarios.sh — Runbook of deliberate failures

```
#!/bin/bash
# Run each scenario, verify behavior matches expected, document results

# SCENARIO 1: Cloud outage during active switchover
```

```
# Kill the cloud API while WattBrain is mid-switchover sequence
docker compose stop api ml          # Cloud goes down
# EXPECTED: WattBrain completes switchover from local inference
# EXPECTED: Relay defaults to fail-safe (GRID) if WattBrain also crashes
# VERIFY:  cat /var/log/wattbrain/relay.log | grep "OFFLINE_MODE"

# SCENARIO 2: InfluxDB disk full (sensor data lost)
docker exec influxdb dd if=/dev/zero of=/tmp/fill bs=1M count=99999
# EXPECTED: Ingestion service logs error, drops telemetry gracefully
# EXPECTED: Ring buffer on WattBrain holds data until space frees
# EXPECTED: Alert fires to ops team via PagerDuty

# SCENARIO 3: Kafka lag spike (sensor data backlog)
docker compose stop kafka
sleep 300 # 5 minutes of data queued in WattBrain buffers
docker compose start kafka
# EXPECTED: Data catches up in order without duplication
# EXPECTED: InfluxDB shows no time-series gap in the fill

# SCENARIO 4: WattBrain process crash during active automation
sudo kill -9 $(pgrep wattbrain-infer) # Kill ML inference mid-cycle
# EXPECTED: BCM2835 watchdog triggers within 30 seconds
# EXPECTED: Relay outputs default to GRID (open)
# EXPECTED: systemd auto-restarts wattbrain-infer within 5 seconds

# SCENARIO 5: Network partition (4G drops for 72+ hours)
sudo iptables -A OUTPUT -d mqtt.wattwise.pk -j DROP
sleep $((72 * 3600)) # Or: manually block for a few hours and extrapolate
sudo iptables -D OUTPUT -d mqtt.wattwise.pk -j DROP
# EXPECTED: SQLite ring buffer holds all 72h of data
# EXPECTED: All data uploads to InfluxDB on reconnect
# EXPECTED: No data loss. Ring buffer must not have overflowed.
```

Chaos Scenarios Checklist

- ☐ **Cloud API down:** WattBrain continues automation from local ONNX model. No remote dependency during operation.
- ☐ **WattBrain process crash:** Watchdog fires, relays go to fail-safe, systemd restarts service within 10 seconds.
- ☐ **72-hour network blackout:** All data held in ring buffer. Zero data loss. Full catch-up on reconnect.
- ☐ **Kafka full / unavailable:** Ingestion service handles gracefully, no crash, alert fires.
- ☐ **Malformed sensor packet:** Ingestion service rejects it, logs it, continues processing valid packets.
- ☐ **JWT expired mid-session:** Frontend silently refreshes via HttpOnly cookie. User sees no interruption.

- ☐ **ML model OOM on edge:** ONNX inference fails gracefully, falls back to schedule-based prediction.
- ☐ **Database connection pool exhausted:** API returns 503 with Retry-After header. No data corruption.
- ☐ **Savings record UPDATE attempted:** PostgreSQL rule silently blocks it. Audit log records the attempt.

ML VALIDATION · CRITICAL FOR BILLING INTEGRITY

ML Model Validation — How to Know the Models Actually Work

GOP Model: Validation Without a Real Factory

Test	How to Run	Pass Criteria
Historical backtest	Train on 2023–2024 FESCO data. Test on 2025 held-out data.	Precision >88%, Recall >80%, False alarm <5%
Digital twin validation	Run digital twin for 30 days simulated. Compare GOP predictions vs twin's actual outages.	Every simulated outage preceded by prediction >0.85 confidence
Frequency sag detection	On home bench: use variac to drop voltage gradually. Check if GOP confidence rises.	GOP confidence increases monotonically as voltage drops toward 380V
Calibration check	When GOP says 0.85 confidence, it should be right ~85% of the time (not 40%, not 99%).	Calibration plot: predicted probability vs observed frequency within ±5%

Prophet Baseline: Validate It's Not Overfitting

```
def test_baseline_is_realistic():
    # Train Prophet on Jan-Oct data, predict Nov
    baseline_nov = prophet_model.predict_november()
    actual_nov    = df_nov['actual_cost_pkr'].sum()

    # If WattWise wasn't deployed: baseline should be within 10% of actual
    error_pct = abs(baseline_nov - actual_nov) / actual_nov
    assert error_pct < 0.10, f"Prophet baseline error {error_pct:.1%} too high!"

    # Critical: baseline MUST NOT be lower than actual cost
```

```
# (would mean we're claiming savings that didn't happen)
assert baseline_nov >= actual_nov * 0.95, "Baseline too low - billing fraud risk!"
```

- **Critical billing integrity test:** If your Prophet baseline underestimates what the factory would have spent, you are charging for savings that didn't happen. This is fraud. Validate that the baseline overestimates slightly (5–10%) rather than underestimates. Build this as a CI/CD gate — if baseline accuracy drops below 90%, block the monthly billing run.

BILLING VALIDATION · REVENUE CRITICAL

Billing Accuracy Validation

Your billing math must be perfect. One Rs. 100,000 billing error that a factory owner discovers will kill your reputation faster than any technical failure. Test this exhaustively with known inputs and expected outputs before billing a single customer.

apps/api/internal/billing/billing_test.go — Savings calculation tests

```
func TestGainShareCalculation(t *testing.T) {
    tests := []struct {
        baseline, actual, expectedFee, expectedNetSaving float64
    }{
        // Standard case: Rs. 18.2M baseline, Rs. 12.94M actual
        {18_200_000, 12_940_000, 1_052_000, 4_208_000},
        // Edge case: actual cost HIGHER than baseline (rare - should not bill)
        {5_000_000, 6_000_000, 0, 0},
        // Edge case: zero consumption (factory closed for Eid)
        {5_000_000, 0, 1_000_000, 4_000_000},
    }
    for _, tt := range tests {
        gross := tt.baseline - tt.actual
        fee := math.Max(0, gross * 0.20)
        netGain := math.Max(0, gross - fee)
        assert.InDelta(t, tt.expectedFee, fee, 1.0) // within Rs. 1
        assert.InDelta(t, tt.expectedNetSaving, netGain, 1.0)
    }
}

func TestAuditHashIsStable(t *testing.T) {
    // Same input data must always produce same SHA-256 hash
    hash1 := computeAuditHash(testSensorData)
    hash2 := computeAuditHash(testSensorData)
```

```
    assert.Equal(t, hash1, hash2)
    assert.Len(t, hash1, 64) // SHA-256 = 64 hex chars
}
```

BUSINESS · NEXT 30 DAYS

Getting Your First Customer — The Exact Playbook



The mindset shift: Stop thinking "I need to find a customer willing to test my product." Start thinking "I'm offering a factory owner a free energy audit worth Rs. 200,000+ in consulting value, in exchange for 30 days of sensor access." That's a completely different pitch — and it's the true value offer.

Week 1: Build Your Outreach Assets

- 1-page Urdu + English one-pager (what you do, what it costs: nothing, what they get: free energy report)
- 2-minute demo video of the live dashboard
- Savings calculator: "Enter your monthly WAPDA bill → See your estimated savings"
- Simple website: wattwise.pk with calculator

Week 2–3: The 3 Channels

- 1. Warm intros:** Email every professor, family contact, LinkedIn connection asking for a textile industry introduction. One email. 200 people.
- 2. APTMA cold outreach:** 20 personalized LinkedIn messages to Faisalabad mill owners/GMs.
- 3. LUMS NIC / PIEAS:** Apply for incubation — they connect you to industrialists.

The 30-Day Free Shadow Monitoring Agreement

This is your entry document. Simple. One page. Have a lawyer review it. Core terms:

- ☐ **WattWise installs:** Non-invasive CT sensors only. No ATS connection. No relay control. No rewiring.
- ☐ **Duration:** 30 days. Equipment removed at end if factory chooses not to continue.
- ☐ **Factory gets:** A detailed energy audit report with exact kWh data, load shedding impact analysis, and estimated monthly savings.
- ☐ **WattWise gets:** Anonymized aggregate data for model training (factory name not shared without consent).

- ☐ **Cost to factory:** Rs. 0. No obligation to continue.
- ☐ **Data ownership:** Factory owns their own data. WattWise has a license to use it only for their own savings calculations.

FUNDRAISING · PARALLEL TRACK

Seed Round Preparation

You have something most Pakistani startups don't: a **complete, deployed, technically serious product**. Your documentation alone is more thorough than 90% of seed-stage pitches. Now package it for investors.

12-Slide Seed Deck Structure

#	Slide	Key Content	Your Proof Point
1	Cover	WattWise · Logo · 1-line thesis	—
2	The Problem	18–22 hrs load shedding · 3.5x diesel cost · Rs. 2,250/hr waste per mill	NEPRA data citations
3	The Solution	8-second SwiftSwitch™ · Rs. 5M/month savings · Zero rewiring	Live dashboard demo video
4	Product Demo	Screenshots / screen recording of live platform	You have this — record it now
5	How It Works	4-layer architecture in simple terms. Non-technical friendly.	Your architecture diagram
6	Business Model	20% gain-share → Rs. 85K/month subscription · 71% gross margin	Your unit economics table
7	Market Size	5,000+ textile mills Punjab · Rs. 80B TAM · Sialkot next	APTMA / PBS data
8	Traction	X shadow pilots · Y days of data · Z factory interest letters	Get these in next 30 days
9	Go to Market	APTMA beachhead · Gain-share removes sales friction	Your GTM doc

10	Financials	36-month to Rs. 867M ARR · Breakeven at 120 customers	Your financial model
11	Team	CTO: Hammad (PIEAS) + Target CEO co-founder	Your background + what you're hiring for
12	The Ask	USD 250K seed · 18 months runway · 20 customers	Use of funds breakdown

Pakistan Seed Investors to Approach

Stage-Appropriate VCs / Angels

- Sarmayacar — Pakistan's most active early-stage VC
- Indus Valley Capital — Seed + Series A
- i2i Ventures — Pre-seed, strong industrial interest
- Invest2Innovate (i2i) — Accelerator + investment
- Meesho angels network via LUMS alumni

Non-Dilutive Grants First

- IGNITE (ICT R&D Fund): Rs. 5–15M grants for tech companies. Apply immediately.
- PITB Startup Package: Free office, AWS credits, mentorship.
- SMEDA energy efficiency grants: For industrial tech startups.
- WB Climate Innovation: Pakistan cohort for energy/climate startups.

WATTWISE · PHASE 2 VALIDATION & POST-BUILD GUIDE

The hardest part wasn't building it. It's shipping it into a factory.

Build the bench this week. Get the shadow pilot next month. ⚡

